Project Proposal: Coke vs. Pepsi Sugar Prediction Survey

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Project Overview

This project aims to conduct a survey to explore two key questions:
1. **Which soda do people prefer: Coke or Pepsi?**
2. **How accurately can people predict the sugar content in these beverages?**

The survey will collect data on participant preferences and their estimates of sugar content in a standard 12 oz can of Coke and Pepsi. Actual sugar values are **39g for Coke** and **41g for Pepsi**.

The survey will be open to participants from **work colleagues, classmates, and LinkedIn connections** to obtain a diverse dataset.

Incentive and Engagement

To encourage participation, respondents will be entered into a **movie ticket giveaway** where the drawing will be randomized:

- **1st Place Winner:** 4 AMC movie tickets.
- **2nd Place Winners:** 2 participants, each receiving 2 AMC tickets.
- **3rd Place Winners:** 2 participants, each receiving 1 AMC ticket.

Survey Design

Proposed Questions:

1. Which do you prefer: Coke or Pepsi? (Coke, Pepsi)
2. How many grams of sugar are in a standard 12 oz can of Coke?
3. How many grams of sugar are in a standard 12 oz can of Pepsi?
4. Which do you think has more sugar? (Coke, Pepsi, Unsure)

The survey will be hosted via **Google Forms** or a similar platform.

Hypotheses

We will test whether soda preference (Coke vs. Pepsi) impacts predictive ability.

- **Null Hypothesis (H_0):** There is **no difference** in prediction accuracy between participants who prefer Coke and those who prefer Pepsi.
- **Alternative Hypothesis (H_1):** There is a **difference** in prediction accuracy between participants who prefer Coke and those who prefer Pepsi.

We will compare the **mean absolute prediction errors** across preference groups using t-tests or non-parametric equivalents.

Data Analysis Plan

Collected data will be analyzed in R and summarized using:

- **Preference Analysis:**
  Bar charts and pie charts of Coke vs. Pepsi preference.
  Chi-square test to check if preferences differ significantly from 50/50.

- **Prediction Accuracy:**
  Calculate the absolute error for each participant and overall ranking.
  Test predictive ability differences between preference groups (t-test or Wilcoxon).
  Boxplots showing distribution of prediction errors.

- **Insights:**
  Analyze whether preference correlates with better predictions for one soda over the other.

Deliverables

1. **Survey Design:** Google Form with integrated incentive announcement.
2. **Data Analysis:** R script to:
   - Rank participants and identify winners.
   - Produce visualizations (bar charts, boxplots).
   - Run statistical tests (Chi-square, t-tests).
3. **Report:** A Quarto document summarizing survey findings, visuals, and winner rankings.

Timeline

- **Week 1:** Finalize survey questions and launch survey on LinkedIn, with colleagues, and in class.
- **Week 2:** Collect responses.
- **Week 3:** Analyze data and determine winners. Prepare a report and share results.
